

Name:

Period:

HW REVIEW FOR CH6B QUIZ (ON TUES/WED) SECTIONS 6.6, 6.7, 6.5. DUE NEXT CLASS PERIOD

$$(f+g)(x) = f(x) + g(x)$$

$$(f-g)(x) = f(x) - g(x)$$

$$(f \times g)(x) = f(x)g(x)$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \text{ where } g(x) \neq 0$$

6.6 Given $f(x) = x^2 + 2x + 4$ and $g(x) = 3x + 2$

a) Find $f(x) + g(x)$ $x^2 + 2x + 4 + 3x + 2$ $\boxed{x^2 + 5x + 6}$	b) Find $f(x) - g(x)$ $x^2 + 2x + 4 - (3x + 2)$ $x^2 + 2x + 4 - 3x - 2$ $\boxed{x^2 - x + 2}$
c) Find $f(x) \cdot g(x)$ $(x^2 + 2x + 4)(3x + 2)$ $3x^3 + 2x^2 + 6x^2 + 4x + 12x + 8$ $\boxed{3x^3 + 8x^2 + 16x + 8}$	d) Find $\frac{f(x)}{g(x)}$ $\frac{x^2 + 2x + 4}{3x + 2}$ where $3x + 2 \neq 0$ $3x \neq -2$ $\boxed{x \neq -2/3}$
Domain: All Real #s	Domain: $(-\infty, -2/3) \cup (-2/3, \infty)$

Interval Notation

Composition of Functions

Substituting a function or its value into **another** function.

$$f(g(x))$$

Second
First



(inside parentheses
always first)

OR

$$f \circ g(x)$$

6.6 Given that $f(x) = 2x - 5$ and $g(x) = x^2 - 3x + 10$, evaluate each of the following:

a) $(f \circ g)(3)$ $f(g(3))$ $g(3) = 3^2 - 3(3) + 10$ $= 10$ $f(10) = 2(10) - 5$ $= 15$	b) $(g \circ f)(5)$ $g(f(5))$ $f(5) = 2(5) - 5 = 5$ $g(5) = 5^2 - 3(5) + 10$ $= 25 - 15 + 10$ $= 20$	c) $(f \circ g)(x)$ $f(g(x))$ $f(x^2 - 3x + 10)$ $= 2(x^2 - 3x + 10) - 5$ $= 2x^2 - 6x + 20 - 5$ $= 2x^2 - 6x + 15$
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6.7 Find the inverse of the function $f(x) = 2x - 3$

1. Replace $f(x)$ with y	$y = 2x - 3$
2. Interchange x & y	$x = 2y - 3$
3. Solve for y	$x + 3 = 2y$ $y = \frac{x+3}{2}$
4. Replace y with $f^{-1}(x)$	$f^{-1}(x) = \frac{x+3}{2}$

6.7 Find the equation of the inverse of each function.

a) $f(x) = \frac{7x}{4} - 5$ $y = \frac{7x}{4} - 5$ $x = \frac{7y}{4} - 5$ $x + 5 = \frac{7y}{4}$ $4x + 20 = 7y$ $y = \frac{4x+20}{7}$ $f^{-1}(x) = \frac{4x+20}{7}$ $= \text{inverse function}$	b) $f(x) = \sqrt[3]{5x-7}$ $y = \sqrt[3]{5x-7}$ $x = \sqrt[3]{5y-7}$ $(x)^3 = (\sqrt[3]{5y-7})^3$ $x^3 = 5y - 7$ $x^3 + 7 = 5y$ $y = \frac{x^3+7}{5} = f^{-1}(x)$
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Inverse operations Undo each other

6.7 Evaluate.

a) $f(f^{-1}(-5)) = -5$	b) $g^{-1}(g(28)) = 28$
c) $g(g^{-1}(\pi)) = \pi$	d) $f^{-1}(f(x)) = x$

6.5 Solving Square Root Equations

Solve each equation. Check for extraneous solutions IF X IS UNDER THE RADICAL AND ALSO NOT!

a) $(x-6)^{\frac{1}{2}} = 4$ $((x-6)^{\frac{1}{2}})^2 = (4)^2$ $x-6 = 16$ $x = 22$	b) $3\sqrt{x+2} - 7 = 5$ Isolate $\sqrt{}$ $3\sqrt{x+2} = 12$ $\sqrt{x+2} = 4$ $(\sqrt{x+2})^2 = (4)^2$ $x+2 = 16$ $x = 14$
c) $x + \sqrt{x+3} = 3$ MUST CHECK ANSWERS Isolate $\sqrt{}$ $\sqrt{x+3} = 3-x$ $(\sqrt{x+3})^2 = (3-x)^2$ $x+3 = (3-x)(3-x)$ $x+3 = 9-6x+x^2$ Set to zero! $0 = x^2 - 7x + 6$	
Check: $0 = (x-6)(x-1)$ $x-6=0 \quad x-1=0$ $x=6 \quad x=1$	

More on Back ☺

1. Find the domain of
- $f(x)$
- .

$$f(x) = \sqrt{6x+6}$$

What's under $\sqrt{\quad}$ must
be positive or zero

$$6x+6 \geq 0$$

$$6x \geq -6$$

$$x \geq -1$$

2. Find the domain of
- $f(x)$
- .

$$f(x) = \frac{2x+2}{4x-12}$$

$$\text{Denom} \neq 0$$

$$4x-12 \neq 0$$

$$4x \neq 12$$

$$x \neq 3$$

4. Find the domain of
- $f(x)$
- .

$$f(x) = \frac{4x-1}{x^2+x-42}$$

$$x^2+x-42 \neq 0$$

$$(x+7)(x-6) \neq 0$$

$$x \neq -7 \quad x \neq 6$$

5. Find the domain of
- $f(x)$
- .

Tricky ∇ $f(x) = \frac{2x+5}{\sqrt{9x+72}}$

$\sqrt{\quad}$ in the
Denom $\Rightarrow$ Needs

BOTH RULES

$$9x+72 > 0$$

$$9x > -72$$

$$x > -8$$